CLA 4 Report: ML Model Deployment Strategies and Model Serving using FastAPI

Dataset: Titanic – Machine Learning from Disaster (Kaggle), using train.csv.

Model: Logistic Regression with preprocessing pipeline (imputation, scaling and one-hot encoding).

Evaluation accuracy: 0.8045

Observations

1. Online vs Batch Inference: Online inference processes one request/record at a time and returns a prediction immediately. Batch inference processes a collection of records together, making it suitable when many predictions are available at once.

2. Streaming vs Batch Inference: Streaming inference handles records continuously as they arrive, one record at a time. Batch inference waits for a group of records and processes them together.

3. Why FastAPI: FastAPI provides a lightweight REST API framework for exposing the trained model through HTTP endpoints. It also provides request validation with Pydantic and interactive Swagger documentation.

4. Why reuse the same trained model: Reusing the same saved pipeline keeps preprocessing and prediction behavior consistent across online, batch, streaming and API inference. It also avoids retraining and prevents differences caused by using separately trained models.

Inference Evidence

Online inference: 1 unseen record processed.

Batch inference: 20 unseen records processed and saved to batch_predictions.csv.

Streaming inference: 10 unseen records processed one at a time with a 1-second delay.

FastAPI: /health and /predict endpoints implemented; Pydantic validation is included.